AGA KHAN UNIVERSITY EXAMINATION BOARD

HIGHER SECONDARY SCHOOL CERTIFICATE

CLASS XI EXAMINATION

APRIL/ MAY 2017

Chemistry Paper II

Time: 2 hours 20 minutes Marks: 55

AKU-EB May Examination 2017

INSTRUCTIONS

Please read the following instructions carefully.

1. Check your name and school information. Sign if it is accurate.

I agree that this is my name and school. **Candidate's signature**

2. RUBRIC. There are THIRTEEN questions. Answer ALL questions. Questions 12 & 13 each offer TWO choices. Attempt any ONE choice from each.

3. When answering the questions:

   Read each question carefully.
   Use a black pointer to write your answers. DO NOT write your answers in pencil.
   Use a black pencil for diagrams. DO NOT use coloured pencils.
   DO NOT use staples, paper clips, glue correcting fluid, or ink erasers.
   Complete your answer in the allocated space only. DO NOT write outside the answer box.

4. The marks for the questions are shown in brackets ( ).

5. You may use a scientific calculator if you wish.

Q.1. (Total 4 Marks)

NH_3 is prepared by heating 3.74 moles of NH_4Cl and 2.7 moles of $Ca(OH)_2$.

$$2NH_4Cl_{(s)} + Ca(OH)_{2(s)} \longrightarrow CaCl_{2(s)} + 2NH_{3(g)} + 2H_2O_{(l)}$$

a. Identify the limiting reactant for the given reaction. (3 Marks)

b. Calculate the amount of NH_3 produced in the given reaction. (1 Mark)

(**Note**: Atomic mass of N = 14 amu, H = 1 amu)

AKU-EB May Examination 2017

Q.2. (Total 5 Marks)

a. State any THREE defects of Bohr's atomic model. (3 Marks)

b.

i. State Hund's rule. (1 Mark)

ii. Give any ONE example showing application of Hund's rule. (1 Mark)

Q.3. (Total 3 Marks)

SiH_4 is a non-polar covalent molecule with polar Si – H bonds.

a. Identify the geometry of SiH_4 molecule. (1 Mark)

b. Why are the Si – H bonds polar? (1 Mark)

c. Why is the SiH_4 molecule non-polar? (1 Mark)

AKU-EB May Examination 2017

Q.4. (Total 3 Marks)

Identify the type of hybridisation, geometry and bond angle between hybrid orbitals of the molecule in the given table.

Molecule	Hybridisation	Geometry	Bond Angle
$BeCl_2$			

Q.5. (Total 4 Marks)

Calculate the mass of 10 dm^3 of oxygen (O_2) gas at 80°C and 3000 mm Hg pressure.

(**Note**: The value of R = 0.0821 dm^3 atm K^{-1} mol^{-1} and atomic mass of O = 16 amu)

AKU-EB May Examination 2017

Q.6. (Total 3 Marks)

a. Why is the rate of evaporation of water less as compared to petrol? (1 Mark)

_____

_____

b. What will be the effect of the following factors on the rate of evaporation of liquids?

i. Surface area (1 Mark)

_____

_____

ii. Temperature (1 Mark)

_____

_____

Q.7. (Total 4 Marks)

a. Define reversible reaction and give ONE example of it. (2 Marks)

_____

_____

_____

_____

b. Predict in which direction the equilibrium will shift in the given situations.

i. $K_c > \frac{[Product]}{[Reactant]}$ (1 Mark)

Direction: _____

ii. $K_c < \frac{[Product]}{[Reactant]}$ (1 Mark)

Direction: _____

AKU-EB May Examination 2017

Q.8. (Total 4 Marks)

a. State Le-Chatelier's principle. (1 Mark)

b. Determine the extent of the following reactions by considering their magnitude of equilibrium constant (K_c) given in the table below. (3 Marks)

i.	$H_{2(g)} + Br_{2(g)} \rightleftharpoons 2HBr_{(g)}$	$K_c = 5.4 \times 10^{18}$
ii.	$N_{2(g)} + O_{2(g)} \rightleftharpoons 2NO_{(g)}$	$K_c = 1.0 \times 10^{-30}$
iii.	$N_2O_{4(g)} \rightleftharpoons 2NO_{2(g)}$	$K_c = 0.36$

i.

ii.

iii.

Q.9. (Total 4 Marks)

On the basis of collision theory, describe the effect of the following factors on the rate of a chemical reaction. Give ONE example in each case.

a. Concentration of reactants (2 Marks)

b. Surface area of reactants (2 Marks)

AKU-EB May Examination 2017

Q.10. (Total 3 Marks)

a. Write ONE difference between osmotic pressure and reverse osmosis. (2 Marks)

Osmotic pressure	Reverse osmosis

b. Give ONE example of reverse osmosis. (1 Mark)

_____

_____

Q.11. (Total 3 Marks)

a. State the first law of thermodynamics. (1 Mark)

_____

_____

b. If 650 J heat is absorbed by a system and 450 J work is done on the system, then find the change in internal energy (ΔE) of the given system. (2 Marks)

_____

_____

_____

_____

AKU-EB May Examination 2017

Q.12. (Total 7 Marks)

EITHER

a.

i. State Heisenberg's uncertainty principle and give its mathematical expression. (2 Marks)

ii. Describe the concept of orbital on the basis of Heisenberg's uncertainty principle. (2 Marks)

iii. Write any THREE differences between orbit and orbital. (3 Marks)

OR

b.

i. Define buffer solution. (1 Mark)

ii. Describe the preparation of a basic buffer. Give ONE example of it. (2 Marks)

iii. An acidic buffer composed of CH_3COOH and CH_3COONa helps to maintain the pH of a solution when an acid or a base is added to it. Justify the given statement with the help of chemical equations. (4 Marks)

AKU-EB May Examination 2017

AKU-EB May Examination 2017

Q.13. (Total 8 Marks)

EITHER

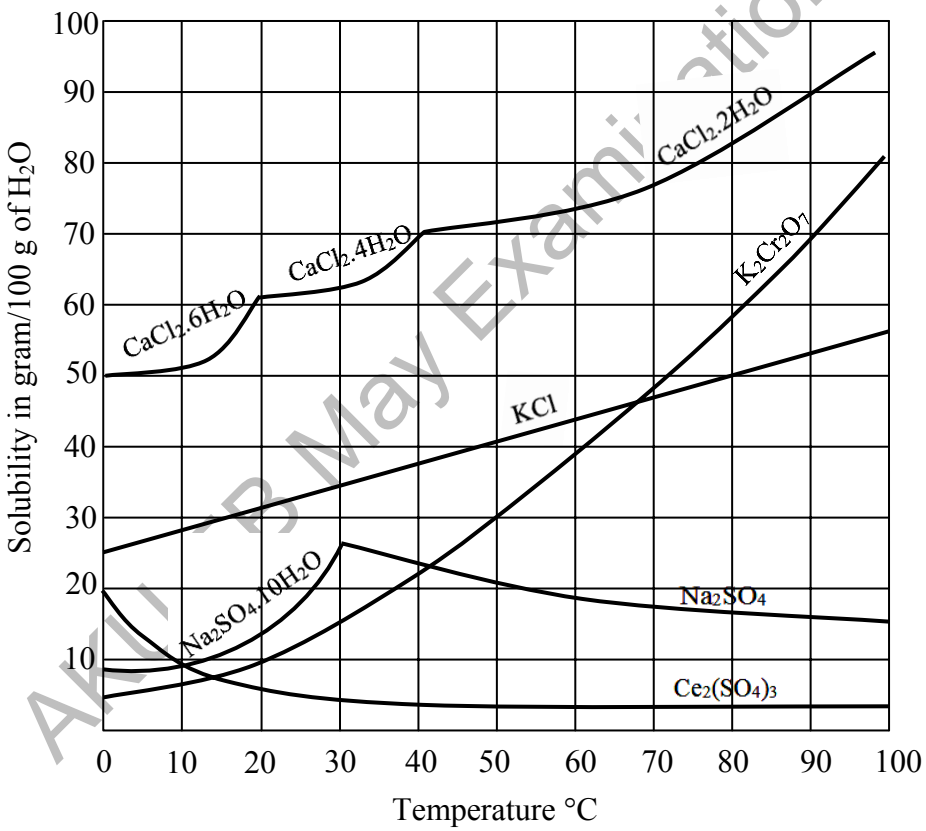
a.

i. Define solubility. Using the given graph, describe in detail the types of solubility curves shown for each salt. (6 Marks)

ii. Deduce the following from the graph. (2 Marks)

I. Grams of undissolved potassium dichromate that settles at the bottom of the beaker when 70 g of it is mixed in 100 g of water at a temperature of 50ºC.

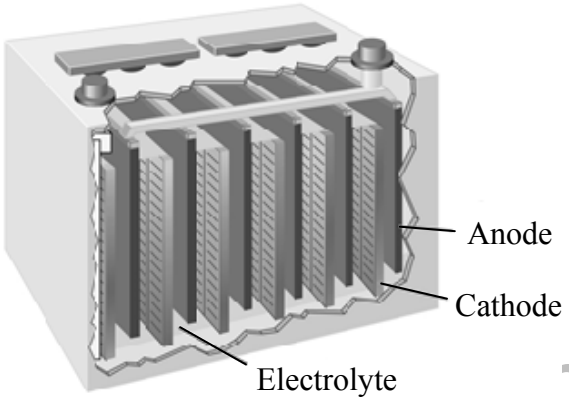
II. Type of potassium chloride solution as saturated or unsaturated or supersaturated when 50 g of it is dissolved in 100 g of water at 80°C.



AKU - EB May Examination 2017

OR

b. The given diagram shows a lead storage battery.



i. Describe the lead storage battery with reference to the following. (5 Marks)

I. Name of anode, cathode and electrolyte
II. Number of cells connected in series to generate 12 volts
III. The most common uses of lead storage battery in daily life

ii. Write balanced equations for the following chemical reactions that occur during the discharge of a lead storage battery. (3 Marks)

I. At the cathode
II. At the anode
III. Overall cell reaction

AKU-EB May Examination 2017

AKU-EB May Examination 2017

END OF PAPER

Please use this page for rough work

AKU-EB May Examination 2017

Please use this page for rough work

AKU-EB May Examination 2017

Please use this page for rough work

AKU-EB May Examination 2017

Please use this page for rough work

AKU-EB May Examination 2017